

TAFLON 25MG FILM COATED TABLETS

Each film-coated tablet contains tenofovir alafenamide fumarate equivalent to 25 mg of tenofovir alafenamide.

WARNING: POSTTREATMENT SEVERE ACUTE EXACERBATION OF HEPATITIS B

Discontinuation of anti-hepatitis B therapy, including Taflon 25mg Film Coated Tablets, may result in severe acute exacerbations of hepatitis B. Hepatic function should be monitored closely with both clinical and laboratory follow-up for at least several months in patients who discontinue anti-hepatitis B therapy, including Taflon 25mg Film Coated Tablets. If appropriate, resumption of anti-hepatitis B therapy may be warranted [see Warnings and Precautions].

1 DESCRIPTION

Yellow, round-shaped, film coated tablets with white or off-white cores.

2 INDICATIONS AND USAGE

Taflon is indicated for the treatment of chronic hepatitis B in adults and adolescents (aged 12 years and older with body weight at least 35 kg) (see section Pharmacodynamic properties).

3 DOSAGE AND ADMINISTRATION

Therapy should be initiated by a physician experienced in the management of chronic hepatitis B.

Posology

Adults and adolescents (aged 12 years and older with body weight at least 35 kg): one tablet once daily.

Treatment discontinuation

Treatment discontinuation may be considered as follows:

- In HBeAg-positive patients without cirrhosis, treatment should be administered for at least 6-12 months after HBe seroconversion (HBeAg loss and HBV DNA loss with anti-HBe detection) is confirmed or until HBs seroconversion or until there is loss of efficacy. Regular reassessment is recommended after treatment discontinuation to detect virological relapse.
- In HBeAg-negative patients without cirrhosis, treatment should be administered at least until HBs seroconversion or until there is evidence of loss of efficacy. With prolonged treatment for more than 2 years, regular reassessment is recommended to confirm that continuing the selected therapy remains appropriate for the patient.

Missed dose

If a dose is missed and less than 18 hours have passed from the time it is usually taken, the patient should take Taflon as soon as possible and then resume their normal dosing schedule. If more than 18 hours have passed from the time it is usually taken, the patient should not take the missed dose and should simply resume the normal dosing schedule.

If the patient vomits within 1 hour of taking Taflon, the patient should take another tablet. If the patient vomits more than 1 hour after taking Taflon, the patient does not need to take another tablet.

Special populations

Elderly

No dose adjustment of Taflon is required in patients aged 65 years and older.

Renal impairment

No dose adjustment of Taflon is required in adults or adolescents (aged at least 12 years and of at least 35 kg body weight) with estimated creatinine clearance (CrCl) ≥ 15 mL/min or in patients with CrCl < 15 mL/min who are receiving haemodialysis.

On days of haemodialysis, Taflon should be administered after completion of haemodialysis treatment.

No dosing recommendations can be given for patients with CrCl < 15 mL/min who are not receiving haemodialysis.

Hepatic impairment

No dose adjustment of Taflon is required in patients with hepatic impairment.

Paediatric population

The safety and efficacy of Taflon in children younger than 12 years of age, or weighing < 35 kg, have not yet been established. No data are available.

Method of administration

Oral administration. Taflon should be taken with food.

4 CONTRAINDICATIONS

Hypersensitivity to the active substance.

5 WARNINGS AND PRECAUTIONS

Severe Acute Exacerbation of Hepatitis B after Discontinuation of Treatment

Discontinuation of anti-hepatitis B therapy, including Taflon 25mg Tablets, may result in severe acute exacerbations of hepatitis B. Patients who discontinue Taflon 25mg Tablets should be closely monitored with both clinical and laboratory follow-up for at least several months after stopping treatment. If appropriate, resumption of anti-hepatitis B therapy may be warranted.

HBV transmission

Patients must be advised that Taflon does not prevent the risk of transmission of HBV to others through sexual contact or contamination with blood. Appropriate precautions must continue to be used.

Patients with decompensated liver disease

There are limited data on the safety and efficacy of Taflon in HBV-infected patients with decompensated liver disease and who have a Child Pugh Turcotte (CPT) score > 9 (i.e. class C). These patients may be at higher risk of experiencing serious hepatic or renal adverse reactions. Therefore, hepatobiliary and renal parameters should be closely monitored in this patient population.

Exacerbation of hepatitis

Flares on treatment

Spontaneous exacerbations in chronic hepatitis B are relatively common and are characterised by transient increases in serum alanine aminotransferase (ALT). After initiating antiviral therapy, serum ALT may increase in some patients. In patients with compensated liver disease, these increases in serum ALT are generally not accompanied by an increase in serum bilirubin concentrations or hepatic decompensation. Patients with cirrhosis may be at a higher risk for hepatic decompensation following hepatitis exacerbation, and therefore should be monitored closely during therapy.

Flares after treatment discontinuation

Acute exacerbation of hepatitis has been reported in patients who have discontinued treatment for hepatitis B, usually in association with rising HBV DNA levels in plasma. The majority of cases are self-limited but severe exacerbations, including fatal outcomes, may occur after discontinuation of treatment for hepatitis B. Hepatic function should be monitored at repeated intervals with both clinical and laboratory follow-up for at least 6 months after discontinuation of treatment for hepatitis B. If appropriate, resumption of hepatitis B therapy may be warranted.

In patients with advanced liver disease or cirrhosis, treatment discontinuation is not recommended since post-treatment exacerbation of hepatitis may lead to hepatic decompensation. Liver flares are especially serious, and sometimes fatal in patients with decompensated liver disease.

Renal impairment

Patients with creatinine clearance < 30 mL/min

The use of Taflon once daily in patients with CrCl ≥ 15 mL/min and < 30 mL/min is based on Week 24 interim data on the efficacy and safety of switching from another antiviral regimen to tenofovir alafenamide in an ongoing reported open-label clinical study of virologically suppressed chronic HBV-infected patients. There are very limited data on the safety and efficacy of Taflon in HBV-infected patients with CrCl < 15 mL/min on chronic haemodialysis.

The use of Taflon is not recommended in patients with CrCl < 15 mL/min who are not receiving haemodialysis.

Nephrotoxicity

A potential risk of nephrotoxicity resulting from chronic exposure to low levels of tenofovir due to dosing with tenofovir alafenamide cannot be excluded.

It is recommended that renal function is assessed in all patients prior to, or when initiating, therapy with Taflon and that it is also monitored during therapy in all patients as clinically appropriate. In patients who develop clinically significant decreases in renal function, or evidence of proximal renal tubulopathy, discontinuation of Taflon should be considered.

Patients co-infected with HBV and hepatitis C or D virus

There are no data on the safety and efficacy of Taflon in patients co-infected with hepatitis C or D virus. Co-administration guidance for the treatment of hepatitis C should be followed.

Hepatitis B and HIV co-infection

HIV antibody testing should be offered to all HBV-infected patients whose HIV-1 infection status is unknown before initiating therapy with Taflon. In patients who are co-infected with HBV and HIV, Taflon should be co-administered with other antiretroviral agents to ensure that the patient receives an appropriate regimen for treatment of HIV.

Co-administration with other medicinal products

Taflon should not be co-administered with medicinal products containing tenofovir alafenamide, tenofovir disoproxil or adefovir dipivoxil.

Co-administration of Taflon with certain anticonvulsants (e.g. carbamazepine, oxcarbazepine, phenobarbital and phenytoin), antimycobacterials (e.g. rifampicin, rifabutin and rifapentine) or St. John's wort, all of which are inducers of P-glycoprotein (P-gp) and may decrease tenofovir alafenamide plasma concentrations, is not recommended.

Co-administration of Taflon with strong inhibitors of P-gp (e.g. itraconazole and ketoconazole) may increase tenofovir alafenamide plasma concentrations. Co-administration is not recommended.

Lactose intolerance

Taflon contains lactose monohydrate. Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicinal product.

6 ADVERSE REACTIONS

The most frequently reported adverse reactions were headache, nausea, and fatigue.

<i>System organ class</i>	
Frequency	Adverse reaction
<i>Nervous system disorders</i>	
Very common	Headache
Common	Dizziness
<i>Gastrointestinal disorders</i>	
Common	Diarrhoea, vomiting, nausea, abdominal pain, abdominal distension, flatulence
<i>Hepatobiliary disorders</i>	
Common	Increased ALT
<i>Skin and subcutaneous tissue disorders</i>	
Common	Rash, pruritus
Uncommon	Angioedema, urticaria
<i>Musculoskeletal and connective tissue disorders</i>	

Common	Arthralgia
<i>General disorders and administration site conditions</i>	
Common	Fatigue

7 DRUG INTERACTIONS

Taflon should not be co-administered with medicinal products containing tenofovir disoproxil, tenofovir alafenamide or adefovir dipivoxil.

Medicinal products that may affect tenofovir alafenamide

Tenofovir alafenamide is transported by P-gp and breast cancer resistance protein (BCRP). Medicinal products that are P-gp inducers (e.g., rifampicin, rifabutin, carbamazepine, phenobarbital or St. John's wort) are expected to decrease plasma concentrations of tenofovir alafenamide, which may lead to loss of therapeutic effect of Taflon.

Co-administration of such medicinal products with Taflon is not recommended.

Co-administration of tenofovir alafenamide with medicinal products that inhibit P-gp and BCRP may increase plasma concentrations of tenofovir alafenamide. Co-administration of strong inhibitors of P-gp with tenofovir alafenamide is not recommended.

Tenofovir alafenamide is a substrate of OATP1B1 and OATP1B3 in vitro. The distribution of tenofovir alafenamide in the body may be affected by the activity of OATP1B1 and/or OATP1B3.

Effect of tenofovir alafenamide on other medicinal products

Tenofovir alafenamide is not an inhibitor of CYP1A2, CYP2B6, CYP2C8, CYP2C9, CYP2C19, or CYP2D6 in vitro. It is not an inhibitor or inducer of CYP3A in vivo.

Tenofovir alafenamide is not an inhibitor of human uridine diphosphate glucuronosyltransferase (UGT) 1A1 in vitro. It is not known whether tenofovir alafenamide is an inhibitor of other UGT enzymes.

8 PREGNANCY, LACTATION AND FERTILITY

Pregnancy

There are no or limited amount of data from the use of tenofovir alafenamide in pregnant women. The use of tenofovir alafenamide may be considered during pregnancy, if necessary.

Lactation

It is not known whether tenofovir alafenamide is secreted in human milk. However, in reported animal studies it has been shown that tenofovir is secreted into milk. There is insufficient information on the effects of tenofovir in newborns/infants.

A risk to the breast-fed newborns/infants cannot be excluded; therefore, tenofovir alafenamide should not be used during breast-feeding.

Fertility

No human data on the effect of tenofovir alafenamide on fertility are available. Animal studies do not indicate harmful effects of tenofovir alafenamide on fertility.

9 EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Taflon has no or negligible influence on the ability to drive and use machines. Patients should be informed that dizziness has been reported during treatment with tenofovir alafenamide.

10 OVERDOSAGE

If overdose occurs the patient must be monitored for evidence of toxicity.

Treatment of overdose with tenofovir alafenamide consists of general supportive measures including monitoring of vital signs as well as observation of the clinical status of the patient.

Tenofovir is efficiently removed by haemodialysis with an extraction coefficient of approximately 54%. It is not known whether tenofovir can be removed by peritoneal dialysis.

11 CLINICAL PHARMACOLOGY

Pharmacodynamic properties

Mechanism of Action

Tenofovir alafenamide is a phosphonamidate prodrug of tenofovir (2'-deoxyadenosine monophosphate analogue). Tenofovir alafenamide enters

primary hepatocytes by passive diffusion and by the hepatic uptake transporters OATP1B1 and OATP1B3. Tenofovir alafenamide is primarily hydrolysed to form tenofovir by carboxylesterase 1 in primary hepatocytes. Intracellular tenofovir is subsequently phosphorylated to the pharmacologically active metabolite tenofovir diphosphate. Tenofovir diphosphate inhibits HBV replication through incorporation into viral DNA by the HBV reverse transcriptase, which results in DNA chain termination.

Tenofovir has activity that is specific to hepatitis B virus and human immunodeficiency virus (HIV-1 and HIV-2). Tenofovir diphosphate is a weak inhibitor of mammalian DNA polymerases that include mitochondrial DNA polymerase γ and there is no evidence of mitochondrial toxicity in vitro based on several assays including mitochondrial DNA analyses.

Antiviral activity

The antiviral activity of tenofovir alafenamide was assessed in HepG2 cells against a panel of HBV clinical isolates representing genotypes A-H. The EC_{50} (50% effective concentration) values for tenofovir alafenamide ranged from 34.7 to 134.4 nM, with an overall mean EC_{50} of 86.6 nM. The CC_{50} (50% cytotoxicity concentration) in HepG2 cells was > 44,400 nM.

Resistance

No amino acid substitutions associated with resistance to Tenofovir alafenamide were identified in these isolates (genotypic and phenotypic analyses).

Cross-resistance

The antiviral activity of tenofovir alafenamide was evaluated against a panel of isolates containing nucleos(t)ide reverse transcriptase inhibitor mutations in HepG2 cells. HBV isolates expressing the rtV173L, rtL180M, and rtM204V/I substitutions associated with resistance to lamivudine remained susceptible to tenofovir alafenamide (< 2-fold change in EC_{50}). HBV isolates expressing the rtL180M, rtM204V plus rtT184G, rtS202G, or rtM250V substitutions associated with resistance to entecavir remained susceptible to tenofovir alafenamide. HBV isolates expressing the rtA181T, rtA181V, or rtN236T single substitutions associated with resistance to adefovir remained susceptible to tenofovir alafenamide; however, the HBV isolate expressing rtA181V plus rtN236T exhibited reduced susceptibility to tenofovir alafenamide (3.7-fold change in EC_{50}). The clinical relevance of these substitutions is not known.

Pharmacokinetics

Absorption

Following oral administration of tenofovir alafenamide under fasted conditions in adult patients with chronic hepatitis B, peak plasma concentrations of tenofovir alafenamide were reported approximately 0.48 hours post-dose. Relative to fasting conditions, the administration of a single dose of tenofovir alafenamide with a high fat meal resulted in an increase in tenofovir alafenamide exposure.

Distribution

The binding of tenofovir alafenamide to human plasma proteins was reported as approximately 80%. The binding of tenofovir to human plasma proteins is less than 0.7% and is independent of concentration.

Biotransformation

Metabolism is a major elimination pathway for tenofovir alafenamide in humans, accounting for > 80% of an oral dose. Tenofovir alafenamide is hydrolysed within cells to form tenofovir (major metabolite), which is phosphorylated to the active metabolite, tenofovir diphosphate. Tenofovir alafenamide is not metabolised by CYP1A2, CYP2C8, CYP2C9, CYP2C19, or CYP2D6. Tenofovir alafenamide is minimally metabolised by CYP3A4.

Elimination

Renal excretion of intact tenofovir alafenamide is a minor pathway with < 1% of the dose eliminated in urine. Tenofovir alafenamide is mainly eliminated following metabolism to tenofovir. Tenofovir is renally eliminated from the body by the kidneys by both glomerular filtration and active tubular secretion.

Linearity/non-linearity

Tenofovir alafenamide exposures are dose proportional over the dose range of 8 to 125 mg.

Pharmacokinetics in special populations

Age, gender and ethnicity

No clinically relevant differences in pharmacokinetics according to age or ethnicity have been identified. Differences in pharmacokinetics according to gender were not considered to be clinically relevant.

Hepatic impairment

In patients with severe hepatic impairment, total plasma concentrations of tenofovir alafenamide and tenofovir are lower than those seen in subjects with normal hepatic function. When corrected for protein binding, unbound (free) plasma concentrations of tenofovir alafenamide in severe hepatic

impairment and normal hepatic function are similar.

Renal impairment

Exposures of tenofovir reported with ESRD (estimated creatinine clearance < 15 mL/min) on chronic haemodialysis who received tenofovir alafenamide were substantially higher than those with normal renal function. No clinically relevant differences in tenofovir alafenamide pharmacokinetics were observed in patients with ESRD on chronic haemodialysis as compared to those with normal renal function.

Paediatric population

No clinically reported relevant differences in tenofovir alafenamide or tenofovir pharmacokinetics were observed between adolescent and adult HIV-1 infected subjects.

12 STORAGE

Preserve in tightly closed container, store not above 30°C at dry place.

13 DOSAGE FORMS AND PACKAGING AVAILABLE

Container of 30 Tablets

14 MANUFACTURED BY

Shijiazhuang Lonzeal Pharmaceuticals Co., Ltd.
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Shenze (No. 16 West Ring Road),
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15 DATE OF REVISION

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